

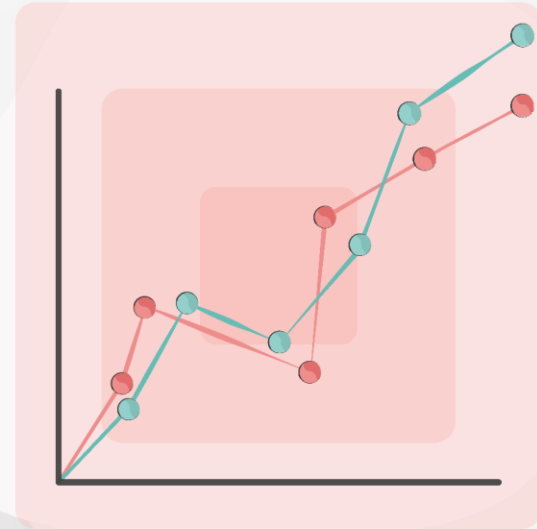


Data engineering

Module: Data pipelines

Topic: Service management

**Welcome to today's
webinar.**



Ice breaker

Discussion...

- How are you feeling today? Motivated, happy, etc.?
- Is there anything about the EPA you are unsure of?
- Think back to a time when a data pipeline you worked on failed or produced unexpected results. What was the cause - and how did you find out?

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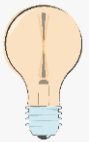
**Submit your responses to the
chat or turn on your
microphone**



The 9 AM dashboard disaster

Inspired by a real incident at a global retailer...

- **Monday morning:** dashboards show blank charts and errors
- **Pipeline ran** - but processed no usable data
- **Root cause:** silent API schema change over the weekend
- **No alerts triggered;** issue discovered too late
- Business decisions made on incomplete insights



This story highlights three key lessons:

1. Pipelines can fail silently.
2. Observability is about more than uptime - it's about correctness.
3. The business impact of data failures is real and immediate.



Image source: [sellersdorsey.com](https://www.sellersdorsey.com)

e-learning recap

Reflecting on your learning...

The e-learning for this topic, covered the following areas:

- **Pipelines need ongoing care** - Service management ensures reliability and trust.
- **Visibility matters** - Monitoring and alerts help catch issues early.
- **Incidents happen** - Respond calmly, communicate clearly, and learn from them.
- **EPA tests more than tech** - It assesses your skills, behaviours, and communication.
- **Prep with purpose** - Reflect, rehearse, and use the STAR method.
- **Clarity counts** - Strong EPA evidence is specific, relevant, and well-explained.



- Do you have any questions about any of these areas?
- Did everything in the e-learning make sense?

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Q&A discussion

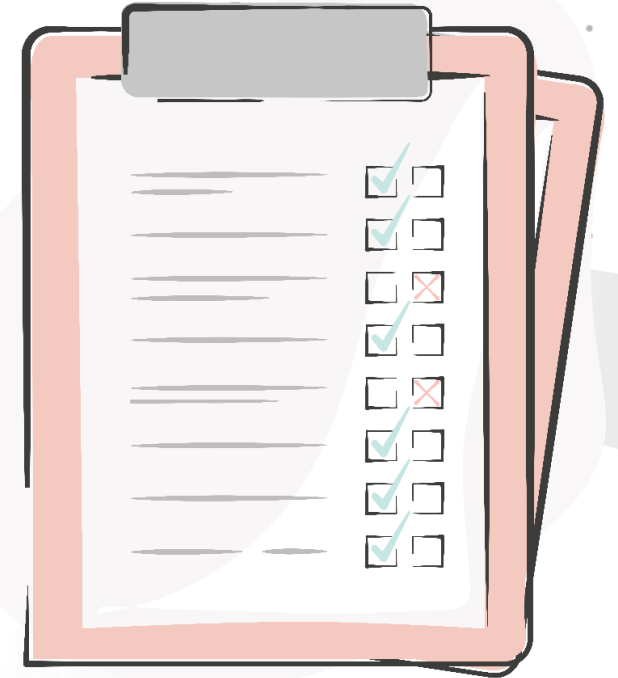
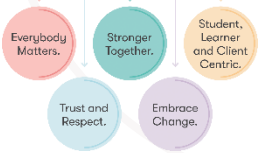


Webinar Agenda

Today, we will cover the following:

1. Welcome and intro
2. Core concept recap
3. Practical lab
4. Summary
5. Q&A

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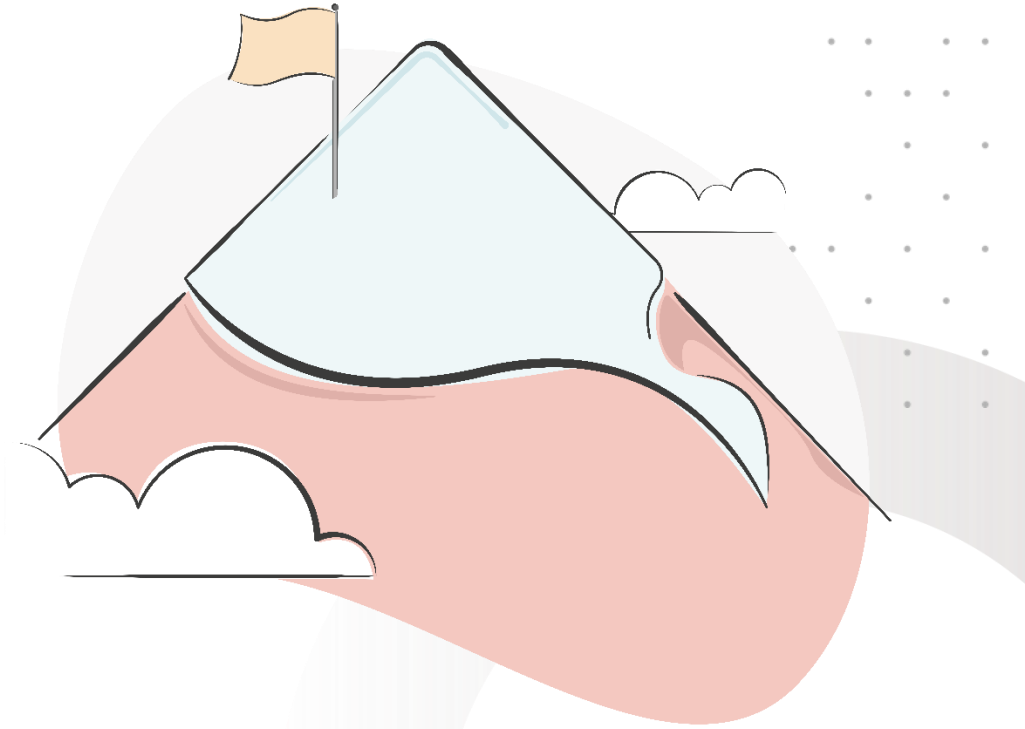


Session aim and objectives

By the end of this session, you should be able to:

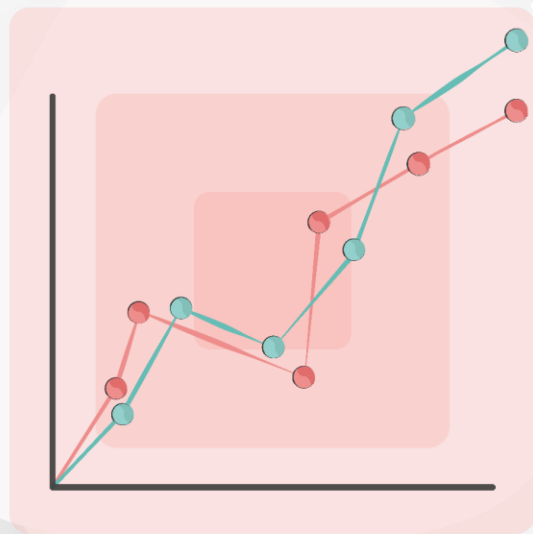
- Understand key service management principles: reliability, availability, observability, and ownership
- Use monitoring and maintenance to keep data pipelines stable and trustworthy
- Handle incidents effectively with clear response and continuous improvement
- Know what the EPA assesses: knowledge, skills, and behaviours (KSBs)
- Prepare for the EPA with reflection, practice, and strong communication
- Learn from real examples to improve your EPA performance

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Core concept recap



Knowledge check poll

Which of the following best describes the principle of observability in service management?

- A. Ensuring that services are always available to users
- B. Automating all data transformations in a pipeline
- C. The ability to monitor, measure, and understand system behaviour
- D. Writing clean and well-documented code

Feedback: C - Observability is about having the tools and data (like logs, metrics, and alerts) to understand what's happening inside your system - especially when something goes wrong.

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**Submit your responses to
the chat!**



What is service management?

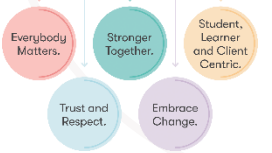
Beyond building pipelines

- Ensures reliability, availability, and trust
- Involves monitoring, maintenance, and ownership
- Keeps data flowing and users confident



- Have you ever seen a pipeline fail silently?
- What happened?

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Ensuring data reliability: The role of Service Management in Keeping Pipelines Flowing Smoothly, image source: www.itgov-docs.com

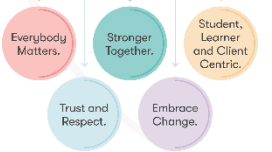
Core principles of service management

The five pillars



The five pillars of service management

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Monitoring & maintenance

Keeping the lights on...

- Track job status, latency, and errors
- Use dashboards, logs, and alerts
- Perform regular checks and updates



- What tools do you use to monitor your pipelines?
- How do you know when something's gone wrong?



Figure: Data Pipeline with Silent Failure

Incident response & recovery

When things break...

- Detect → Diagnose → Resolve
- Communicate clearly and quickly
- Learn from post-incident reviews



What's your process when a job fails?

Who do you notify?

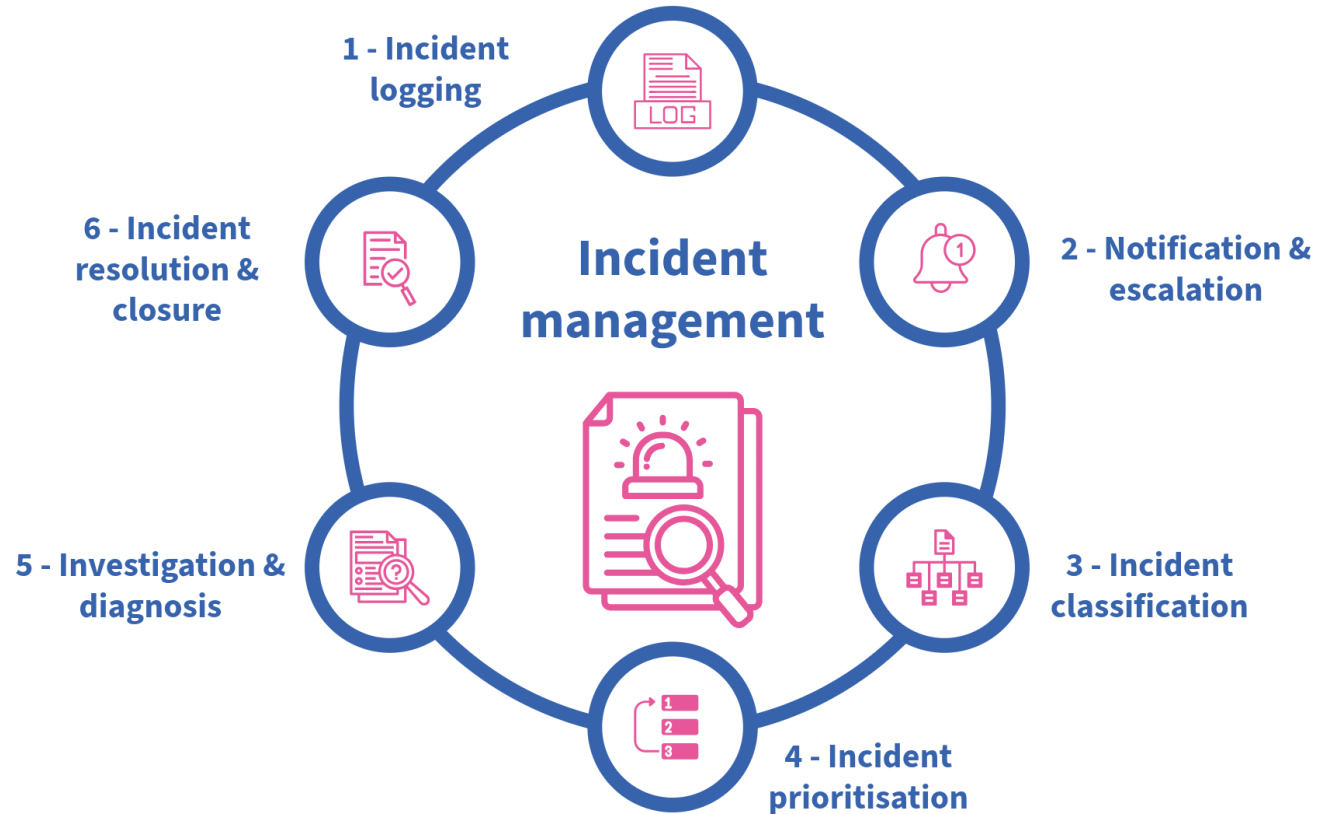
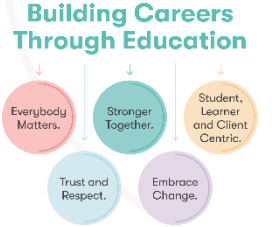
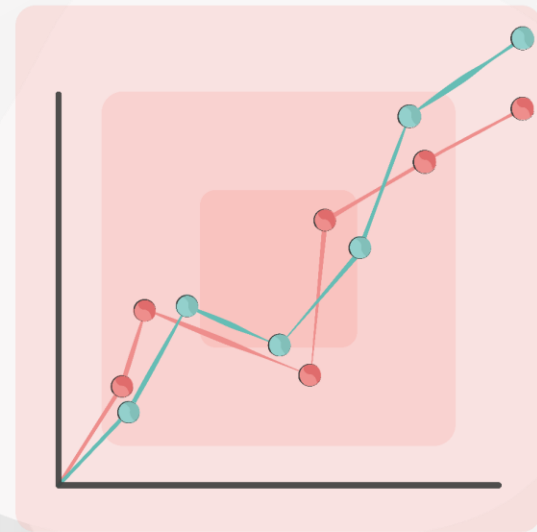


Figure: The Incident Management Process





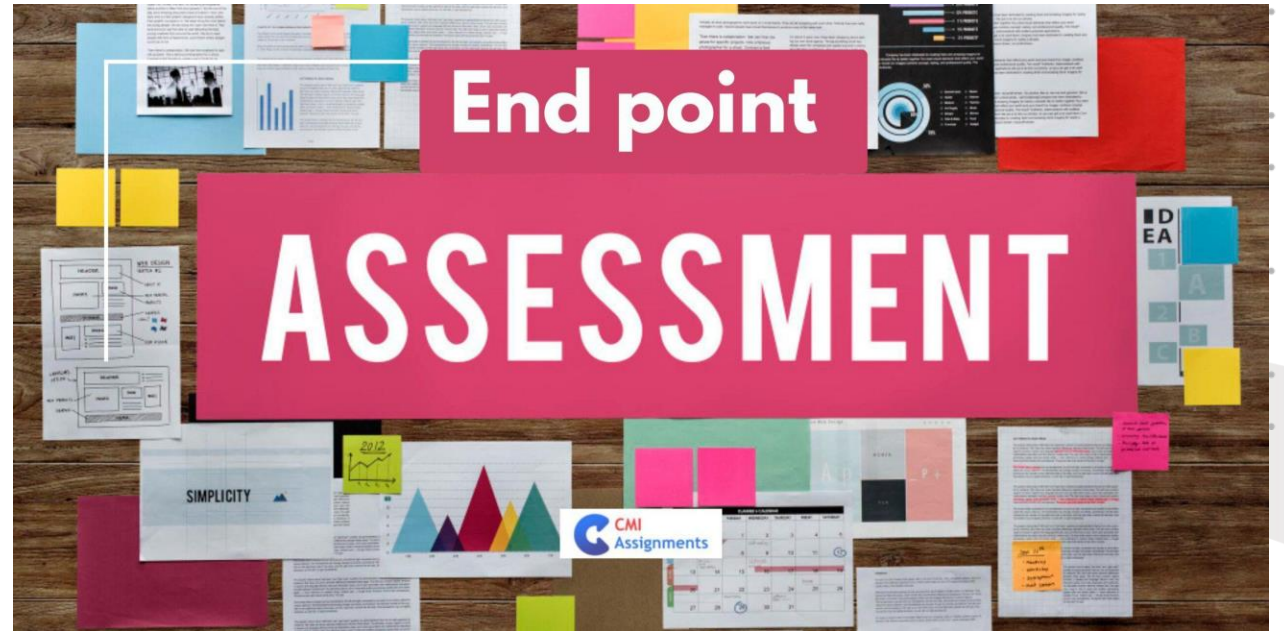
The EPA



Understanding the EPA

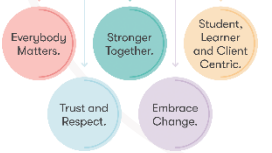
Navigating your academic milestone...

1. Project presentation with Q&A
2. Professional discussion with portfolio
3. Assesses Knowledge, Skills, and Behaviours (KSBs)



Navigating your EPA: What to expect and how to prepare for success, **image source:** cmiassignmenthelp.co.uk

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What the EPA looks for

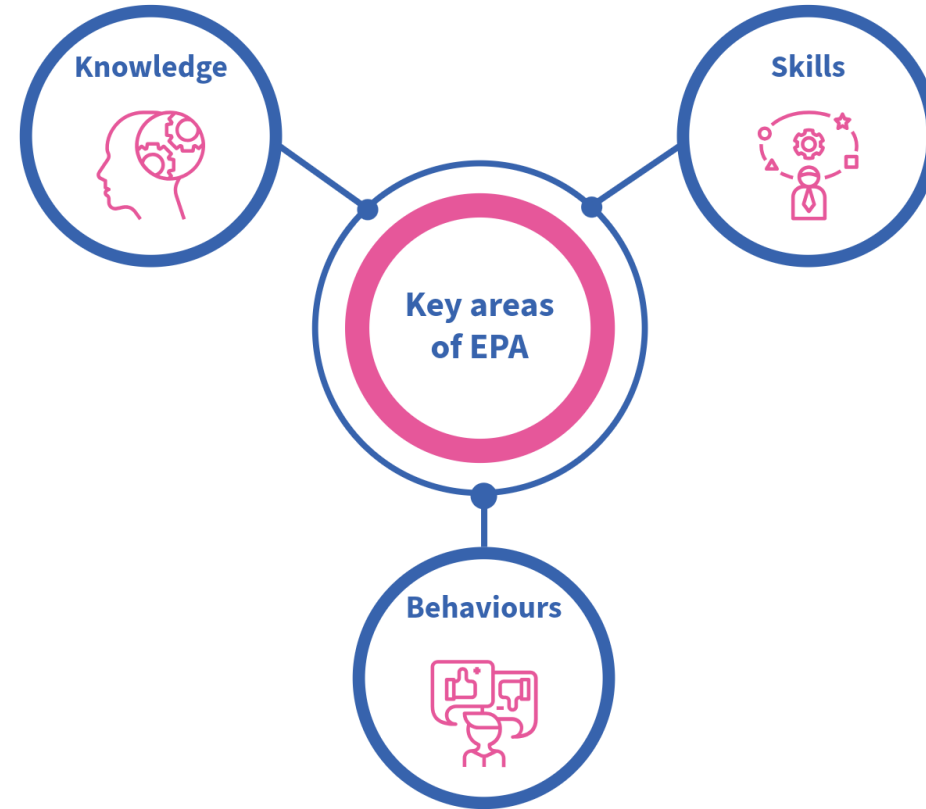
The KSB framework...

1. **Knowledge:** Data modelling, security, quality
2. **Skills:** Pipelines, troubleshooting, collaboration
3. **Behaviours:** Ownership, adaptability, communication

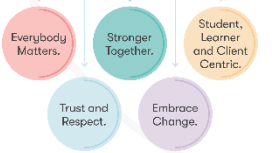


Which of these behaviours do you find easiest to demonstrate?

Which are harder?



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Key Areas of the EPA: Demonstrating Knowledge, Skills, and Behaviours with Confidence



Preparing with purpose

Reflect, rehearse, refine...

- Use the STAR method (Situation, Task, Action, Result)
- Practise explaining your work aloud
- Get feedback from mentors or peers



Have you tried explaining your project to a non-technical friend?

What did you learn?

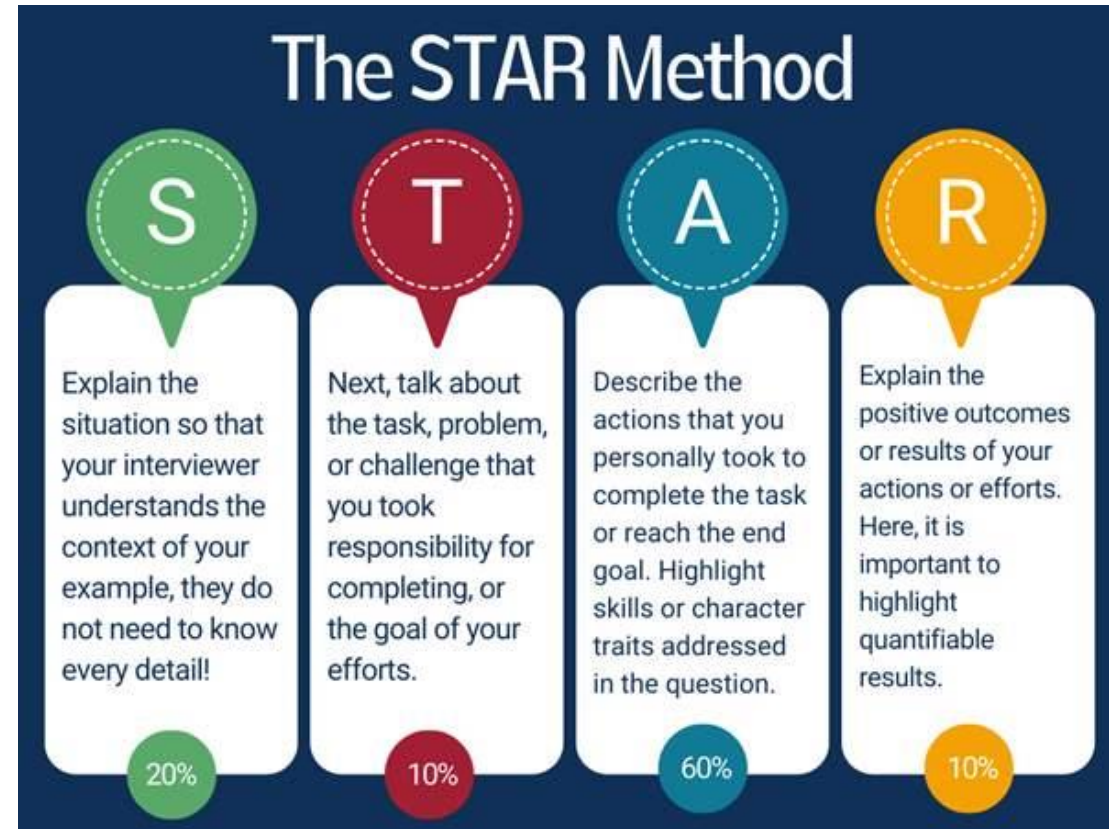
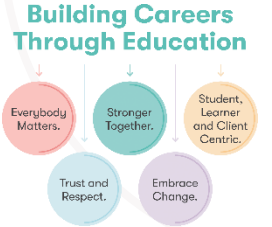


Figure: The STAR method, with percentages representing the time to dedicate to each section of your story. **Image source:** MIT CAPD



What good evidence looks like

Clarity over complexity...

- Be specific and outcome-focused
- Show learning, not just tasks
- Align examples with KSBs
- Even simple projects can shine
- Use the STAR method to structure your story



Strong

"I implemented a data validation layer using Python and Pandas, reducing report errors by 40%. I collaborated with QA and iterated weekly based on feedback."

Vs



Weak

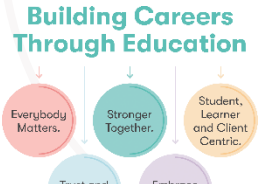
"I worked on a database and fixed some issues. It took a while."

Demonstrating Strong Evidence: Focus on Specific Outcomes and Learning Using the STAR Method



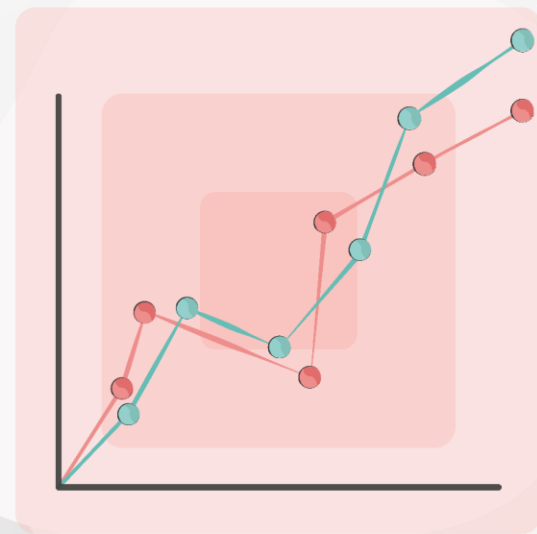
What makes the first example stronger?

What's missing from the second?





Practical lab



Exercise part 5

Next phase of design based on Canadian data

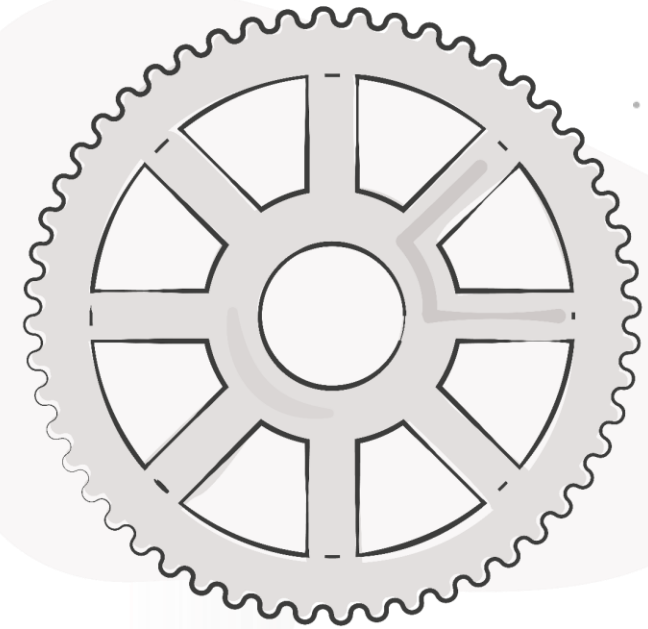
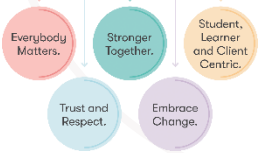
Your step-by-step tasks are as follows:

1. **Update Database Schema:** Modify schema for Canadian data
2. **Re-implement Pipeline:** Clean, validate, and load data sets
3. **Use Provided Test Data:** Data for 10 Canadian users
4. **Login timestamps**

Files Provided in the Hub:

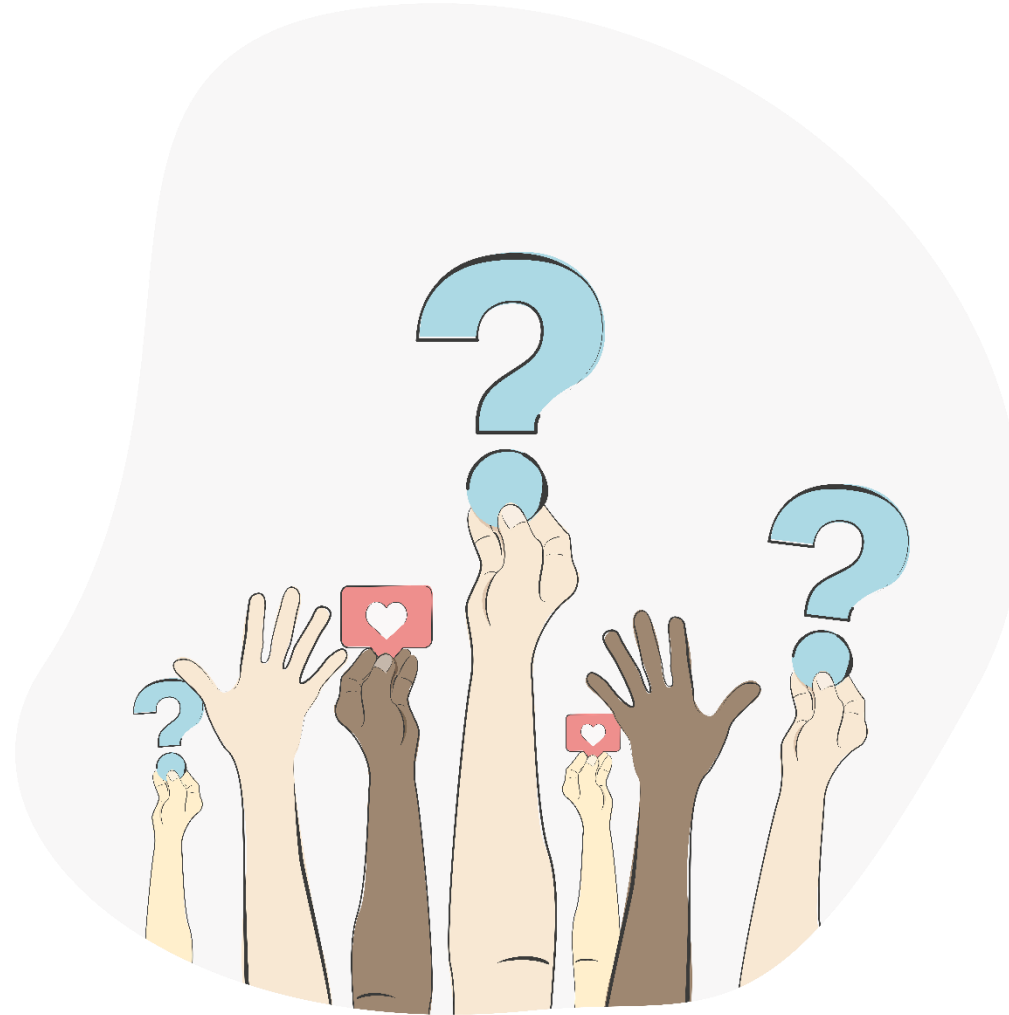
- The 10 sample records from the Canadian subsidiary are in the *CN User Data.csv* file.
- The audit file showing the logins for these 10 users in January 2025 is called *CN-User-LoginTS.csv*

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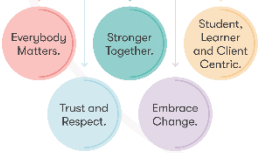


Practical challenge

Any questions or feedback?



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Thank you

